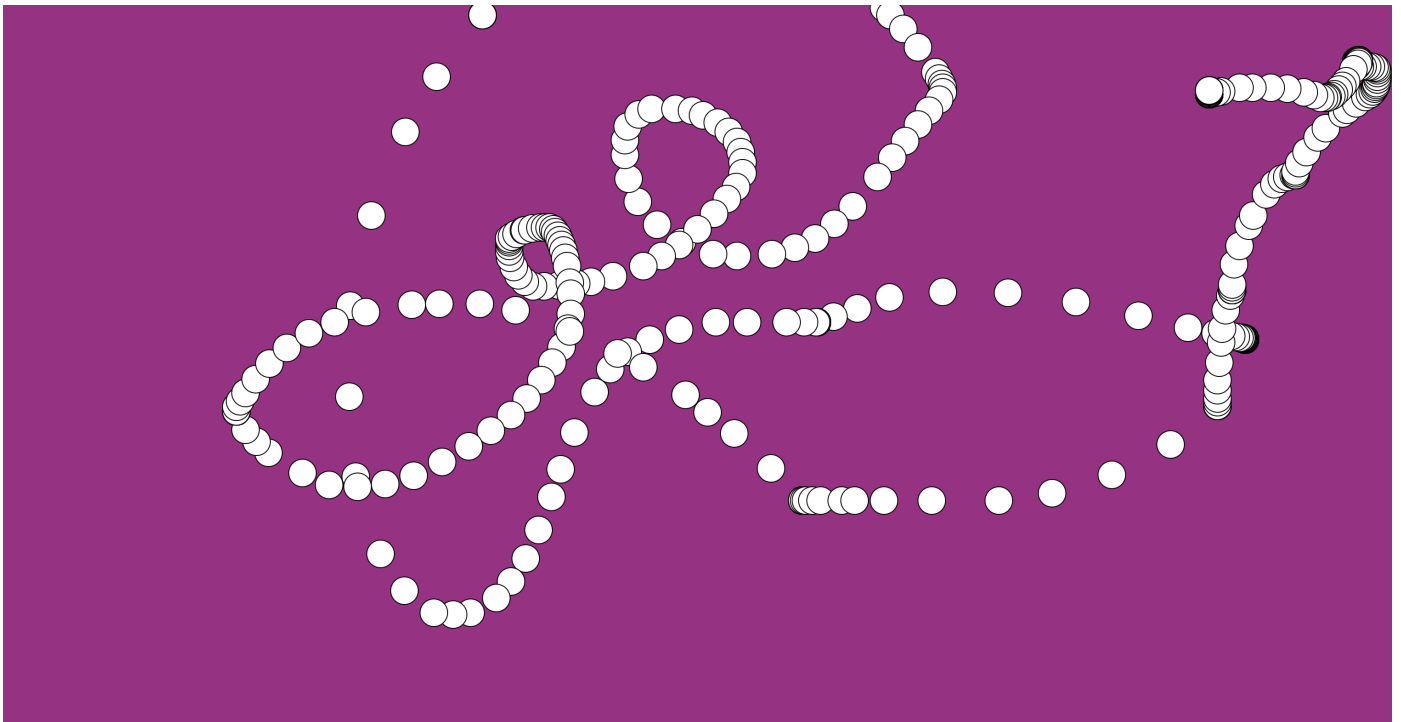


# Week 1

## Day 1

```
function setup() {  
  createCanvas(windowWidth, windowHeight)  
  background(150,50,130)  
  
}  
  
function draw() {  
  // paint program – very basic  
  ellipse(mouseX, mouseY, 30)  
  
}
```



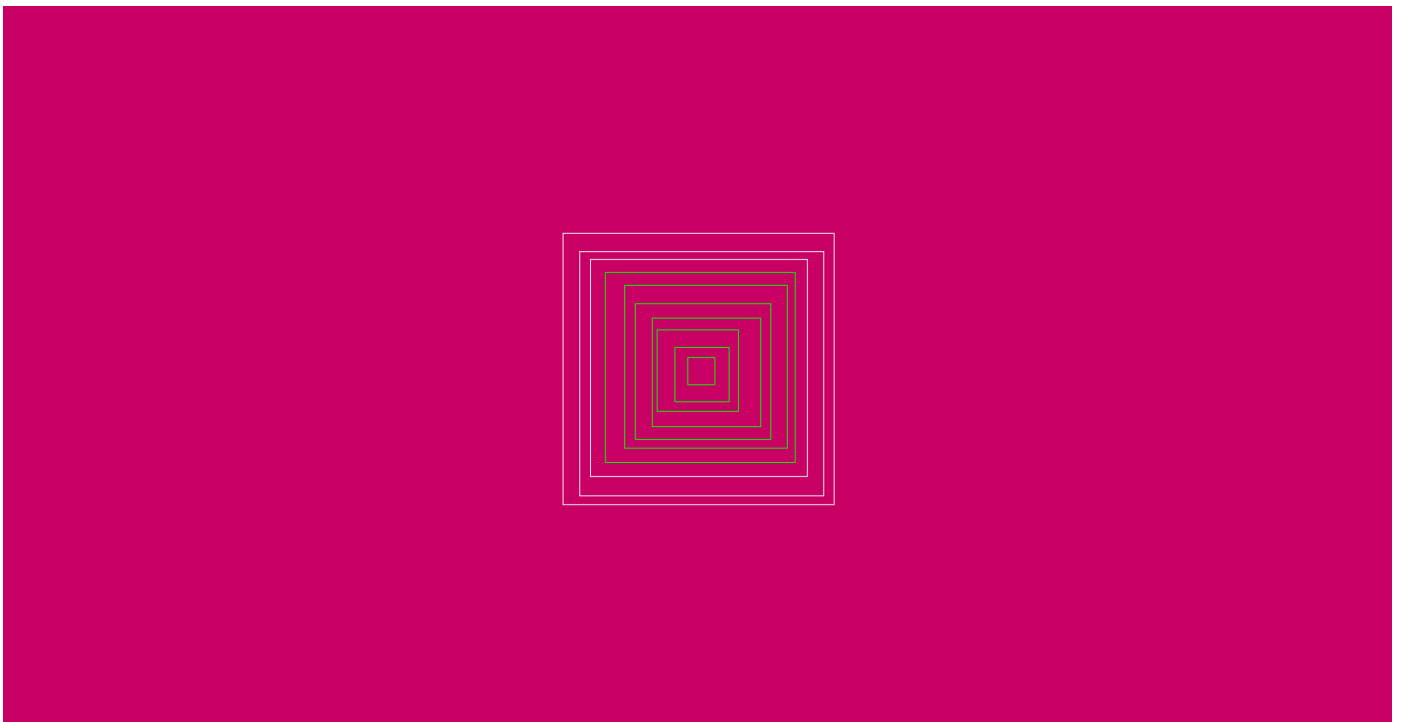
## Day 2

```
let dimX = 400  
let dimY = dimX  
let num = 10  
let reduction = dimX / num  
let posX = 0  
let posY = 0  
  
function setup() {  
  createCanvas(windowWidth, windowHeight)  
  background(200, 0, 100)  
  rectMode(CENTER)  
  angleMode(DEGREES)  
  posX = width / 2  
  posY = height / 2  
}
```

```

function draw() {
  //
  background(200, 0, 100)
  noFill()
  strokeWeight(1)
  stroke(0)
  //animation
  //calculate the dimension of the square
  // based on a sin function
  dimX = 200 + sin(frameCount * 3) * 100
  num = 10
  reduction = dimX / num
  // for loop
  for(let i = 0; i < num ; i++){
    let offsetX = random(10)
    let offsetY = random(10)
    stroke(255)
    if(i > 2){
      stroke(0, 255, 0)
    }
    square(
      posX + offsetX,
      posY + offsetY,
      (dimX) - (reduction * i)
    )
  }
  //text(frameCount, 400, 400)
}

```



```

let dimX = 400
let dimY = dimX
let num = 10
let reduction = dimX / num
let posX = 0

```

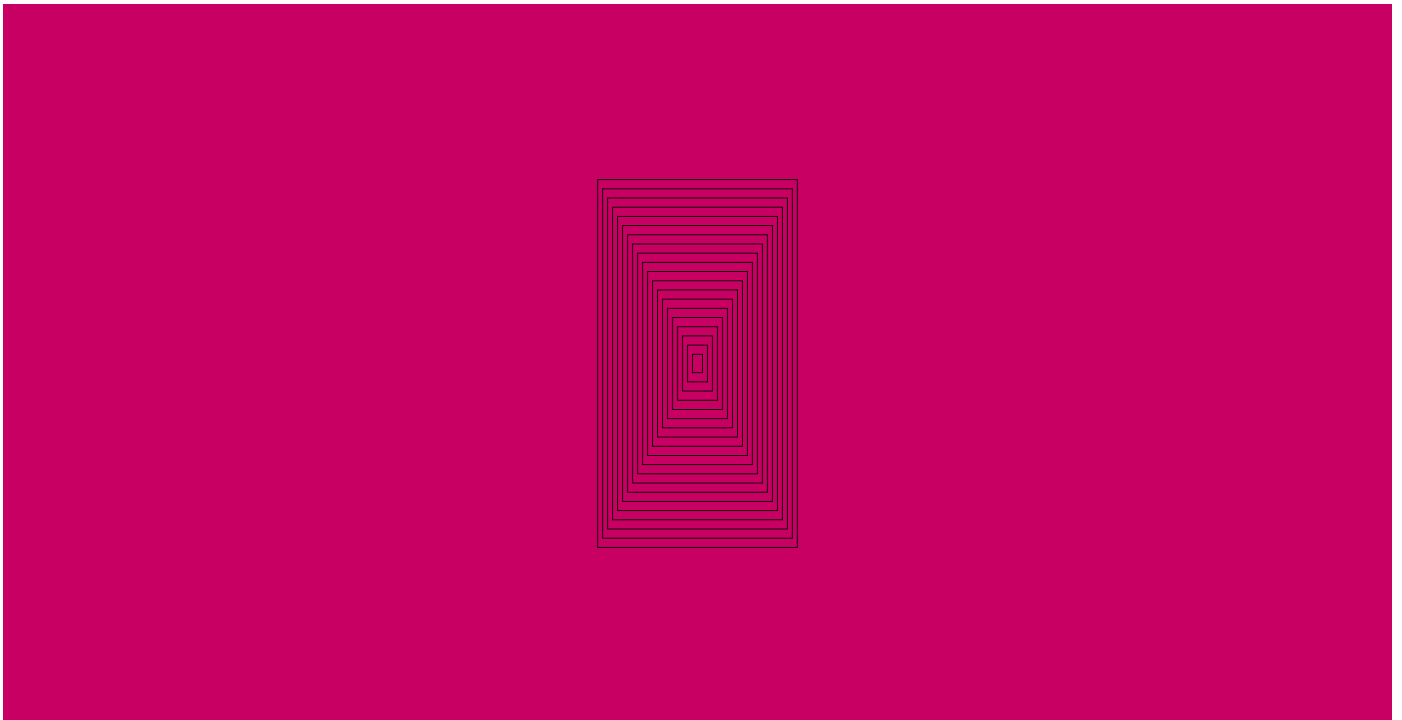
```

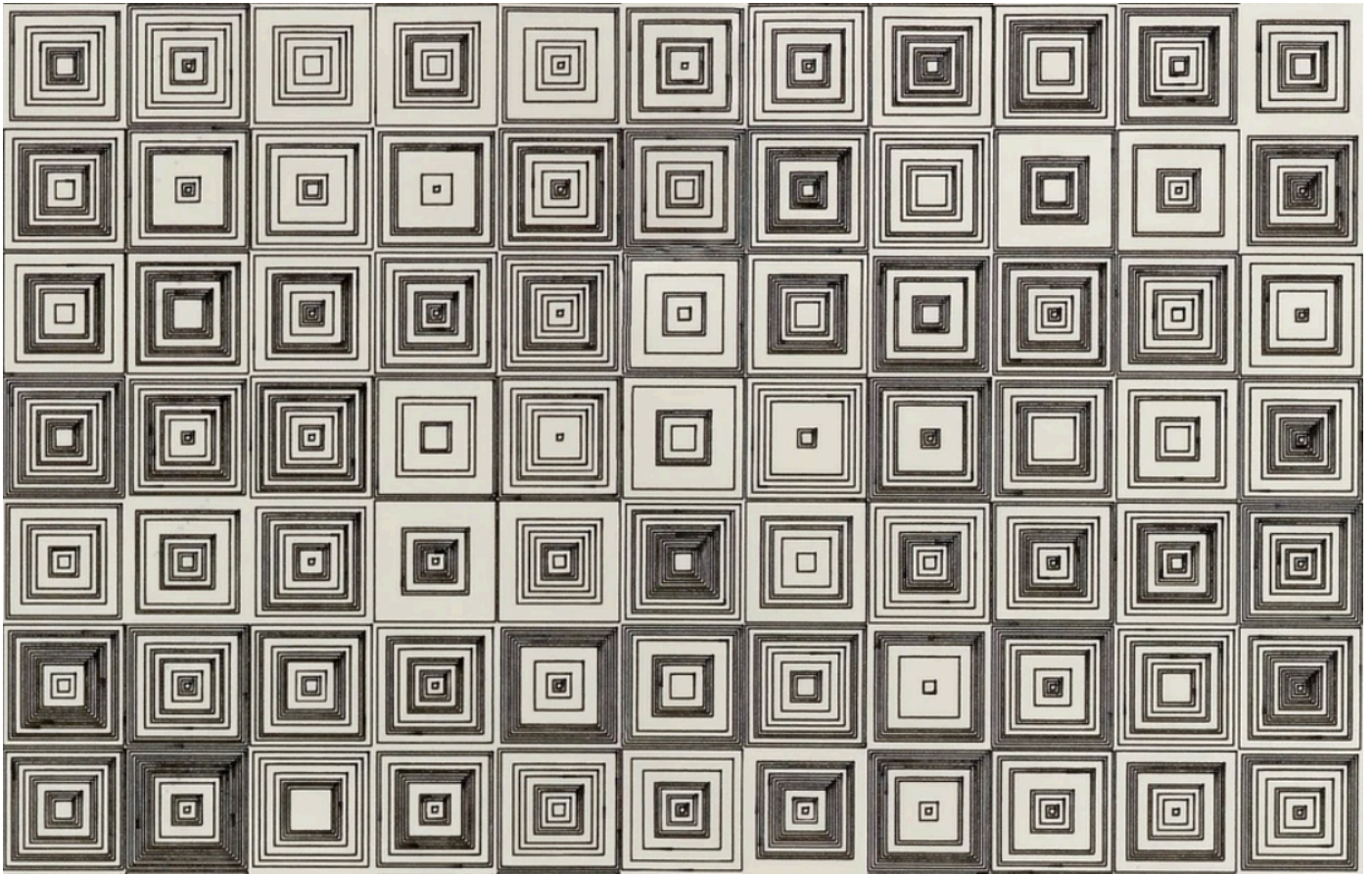
let posY = 0

function setup() {
  createCanvas(windowWidth, windowHeight)
  background(200, 0, 100)
  rectMode(CENTER)
  angleMode(DEGREES)
  posX = width / 2
  posY = height / 2
}

function draw() {
  //
  background(200, 0, 100)
  noFill()
  strokeWeight(1)
  stroke(0)
  dimX = 200 + (sin(frameCount * 5) * 200)
  num = 20
  reduction = dimX / num
  // for loop
  for(let i = 0; i < num ; i++){
    rect(posX, posY,
      (dimX) - (reduction * i),
      (dimY) - (dimY / num * i)
    )
  }
  //text(frameCount, 400, 400)
}

```





## Day 3

```
let dimX = 453
let dimY = dimX
let num = 17
let reduction = dimX / num
let posX = 0
let posY = 0

function setup() {
  createCanvas(windowWidth, windowHeight)
  background(0)
  rectMode(CENTER)
  angleMode(DEGREES)
  posX = width / 2
  posY = height / 2
}

function draw() {
  background(0)
  noFill()
  strokeWeight(3)
  stroke(0)
  let numX = 10
  for(let i = 0; i < numX; i++){
    let dimension = width / numX
    let posX = dimension / 2 + (i * dimension)
    for(let j = 0; j < 5; j++){
      let posY = dimension / 2 + (j * dimension)
      tmcs(posX, posY, 4, dimension, 10)
    }
  }
}
```



```

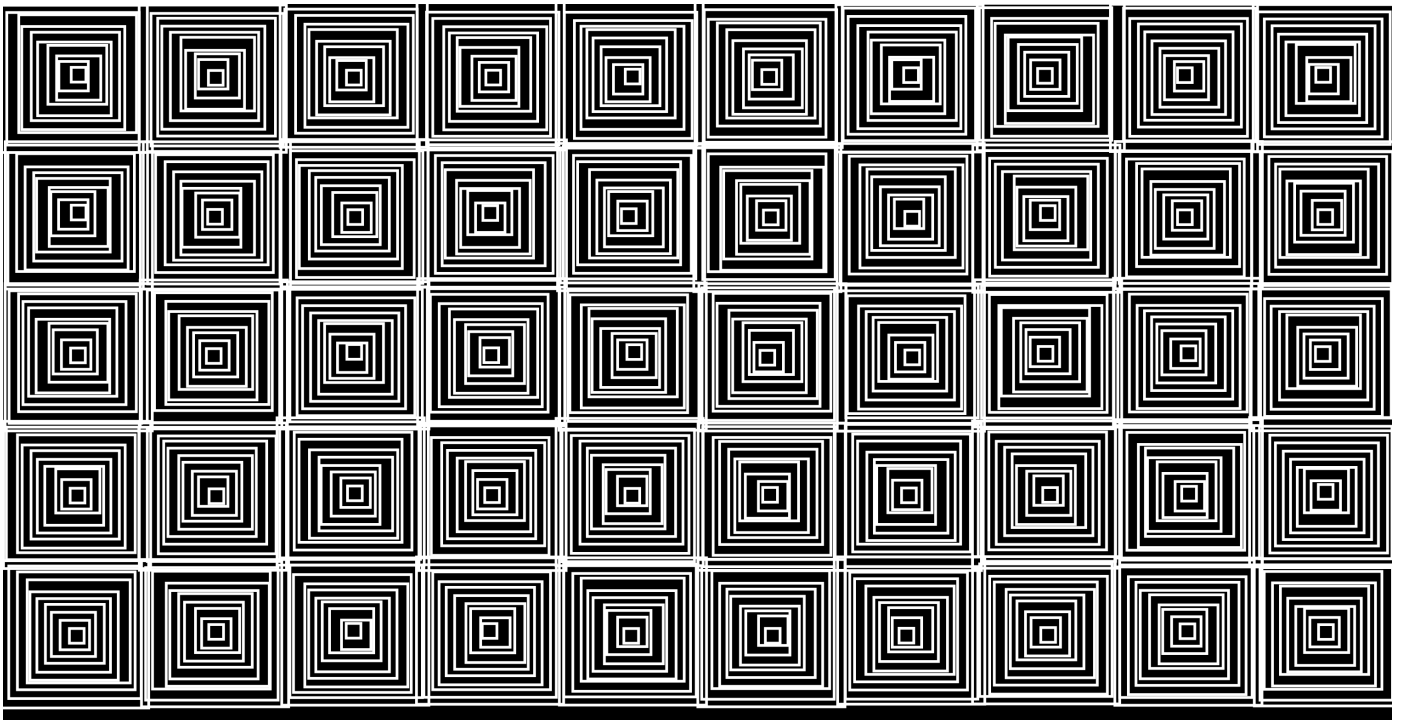
    }
  }

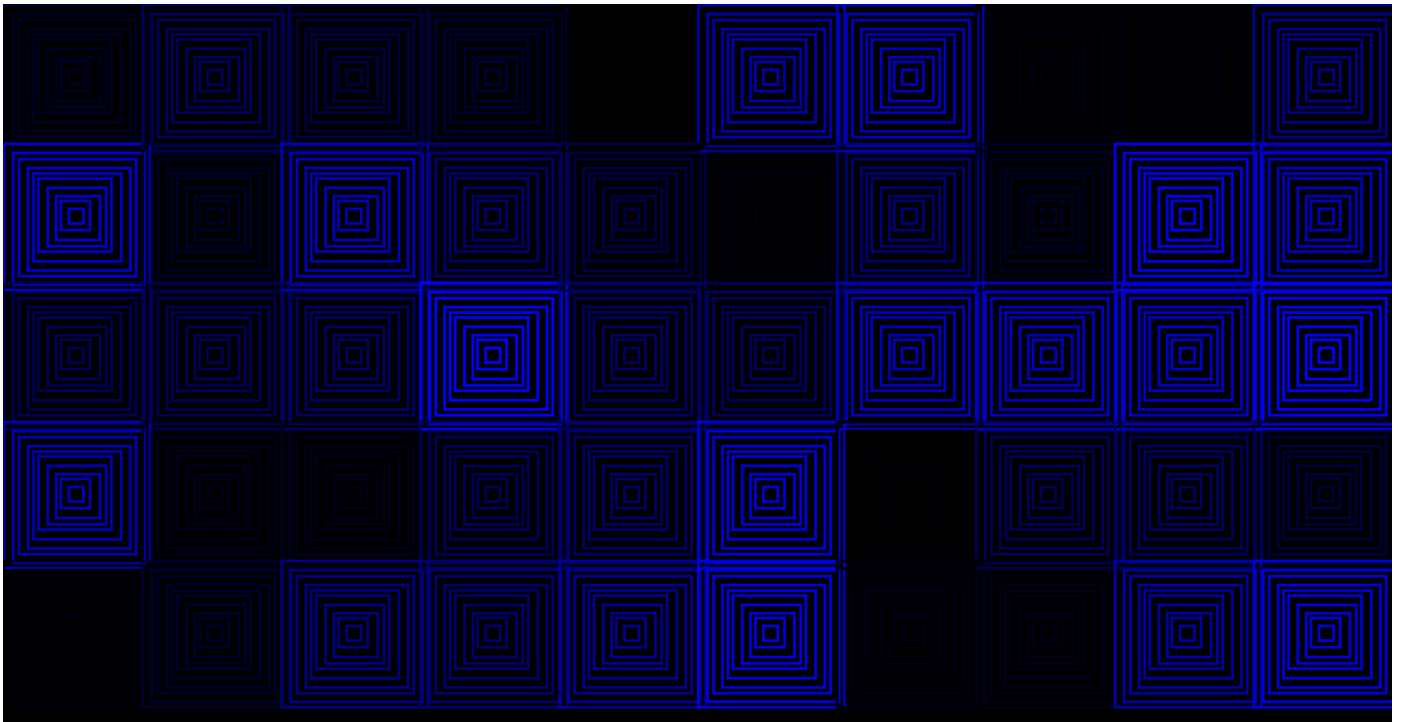
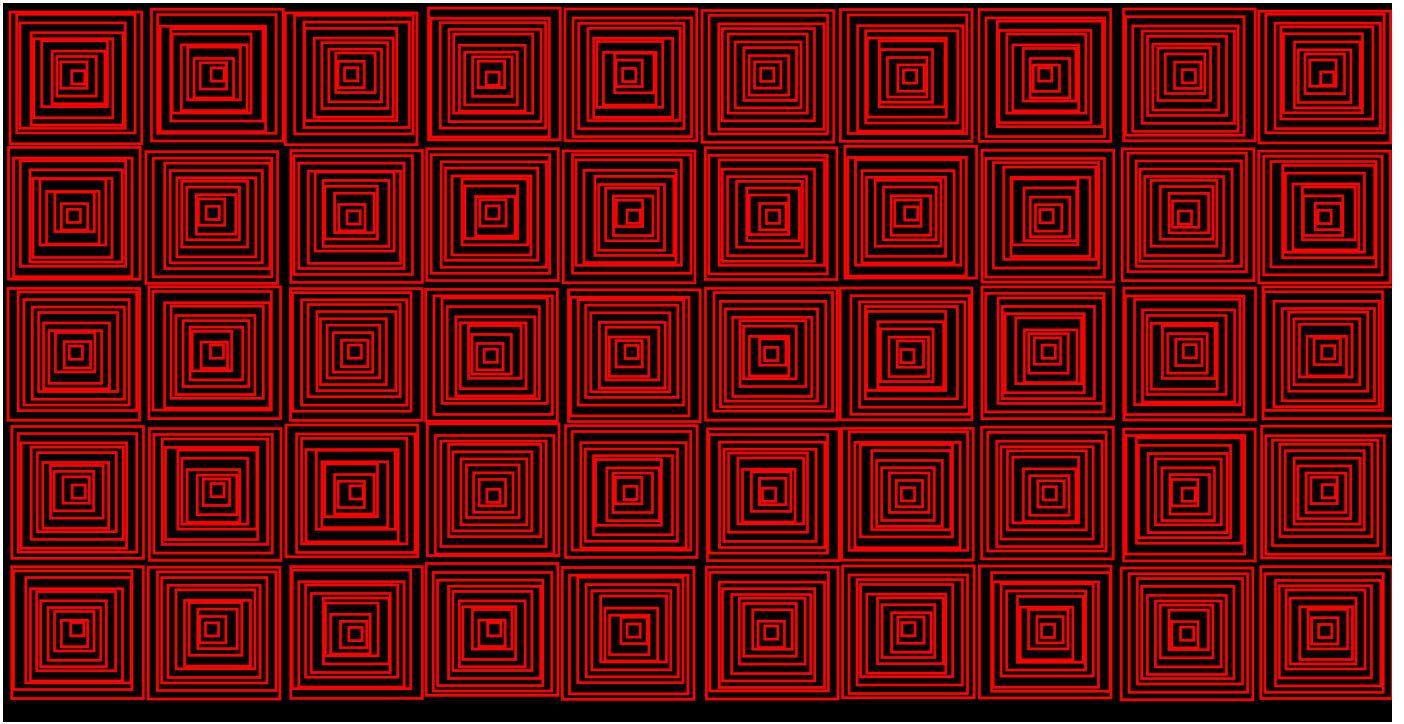
  // this function draws squares at position
  // x and y, and they move randomly
  // a tiny bit
  function tmcs(x, y, speed, dim, num){

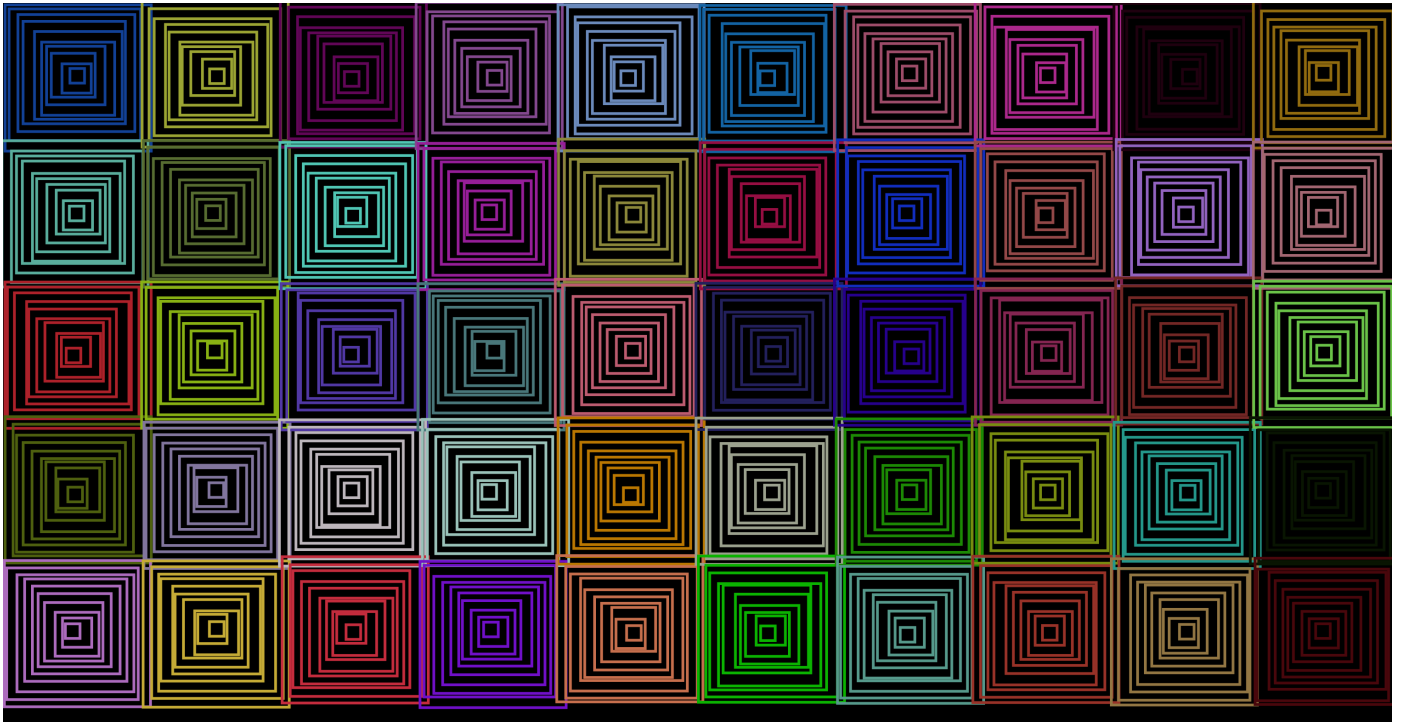
    let dimension = dim + sin(frameCount * speed) * 10
    // num = 10
    let reduction = dimension / num

    for(let i = 0; i < num; i++) {
      let offsetX = random(7)
      let offsetY = random(7)
      stroke(255)
      strokeWeight(3)
      square(
        x+offsetX,
        y+offsetY,
        (dimension) - (reduction * i)
      )
    }
  }
}

```







```
let posX = 0
let posY = 0
let boxDim = 100
function setup() {
  createCanvas(windowWidth, windowHeight, WEBGL)
  background(255)
}

function draw() {
  background(0,0,255)
  orbitControl()
  fill(255)

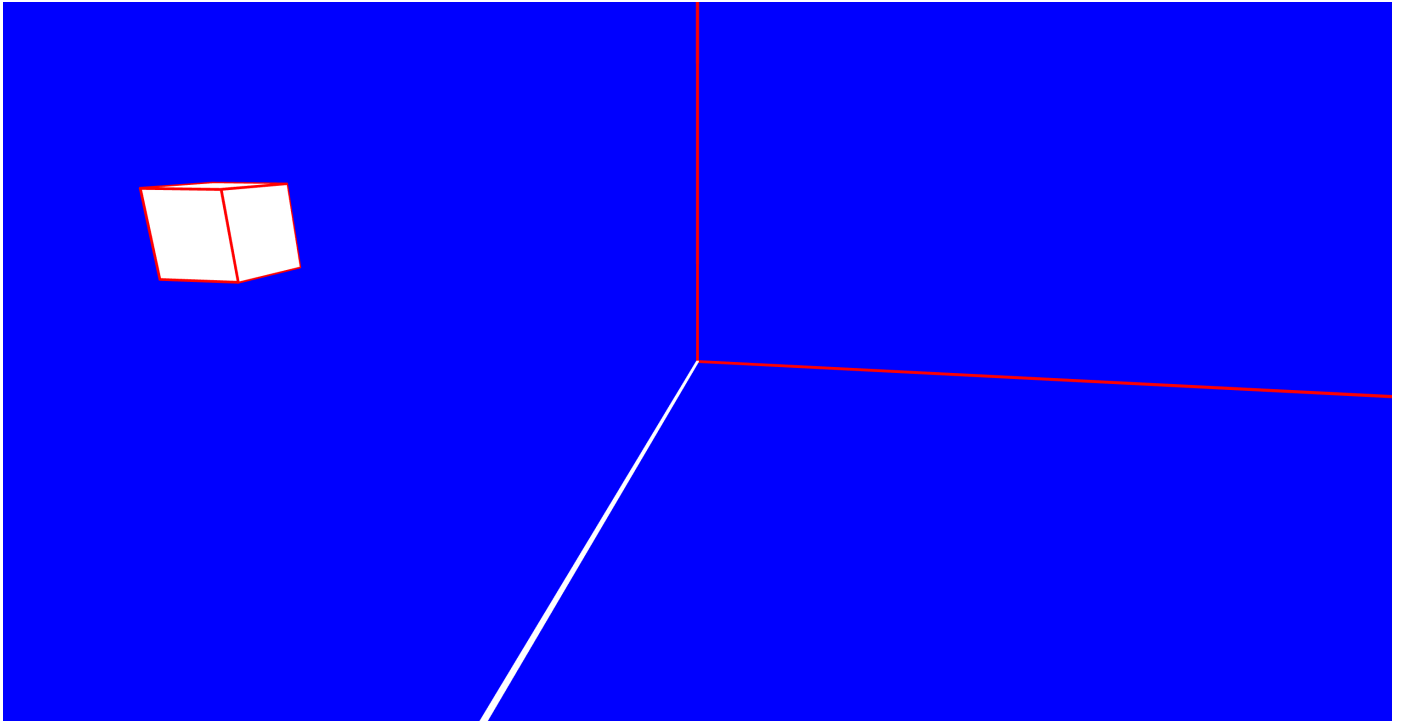
  posX+= 5
  posY = posY - 5
  if(posX > (width / 2)+(boxDim / 2)){
    //posX = 0
    posX -= width/2
  }

  if(posY < -(height / 2)){
    posY = 0
  }

  push()
  translate(posX, posY, 0)
  box(boxDim)
  pop()
  stroke(0)

  stroke(255,0,0)
  strokeWeight(3)
  push()
  line(0,0,0, width, 0, 0)
  line(0,0,0,0,-height,0)
```

```
stroke(255)
line(0,0,0,0,0,1000)
pop()
}
```



```
let x;
let y;
let z;
let xspeed;
let yspeed;
let zspeed;
let depth = 2000;
let dvd;
let r, g, b;

function setup() {
  createCanvas(windowWidth, windowHeight, WEBGL);
  //initialize the variables
  //for the position of the cube
  //at the beginning
  x = random(width);
  y = random(height);
  z = random(-depth, 0);
  //initialize speed at which the cube moves
  xspeed = 4;
  yspeed = 4;
  pickcolor();
}

function pickcolor() {
  r = random(255);
  g = random(255);
  b = random(255);
}
```

```

}

function draw() {
  translate(-width/2, -height/2)
  background(0);
  //noStroke(255);
  //fill(255);
  fill(r, g, b);
  push()
  translate(x,y,z)
  box(100)
  pop()
  //rect(x, y, 80, 60);
  //image(dvd, x, y, 80, 60);

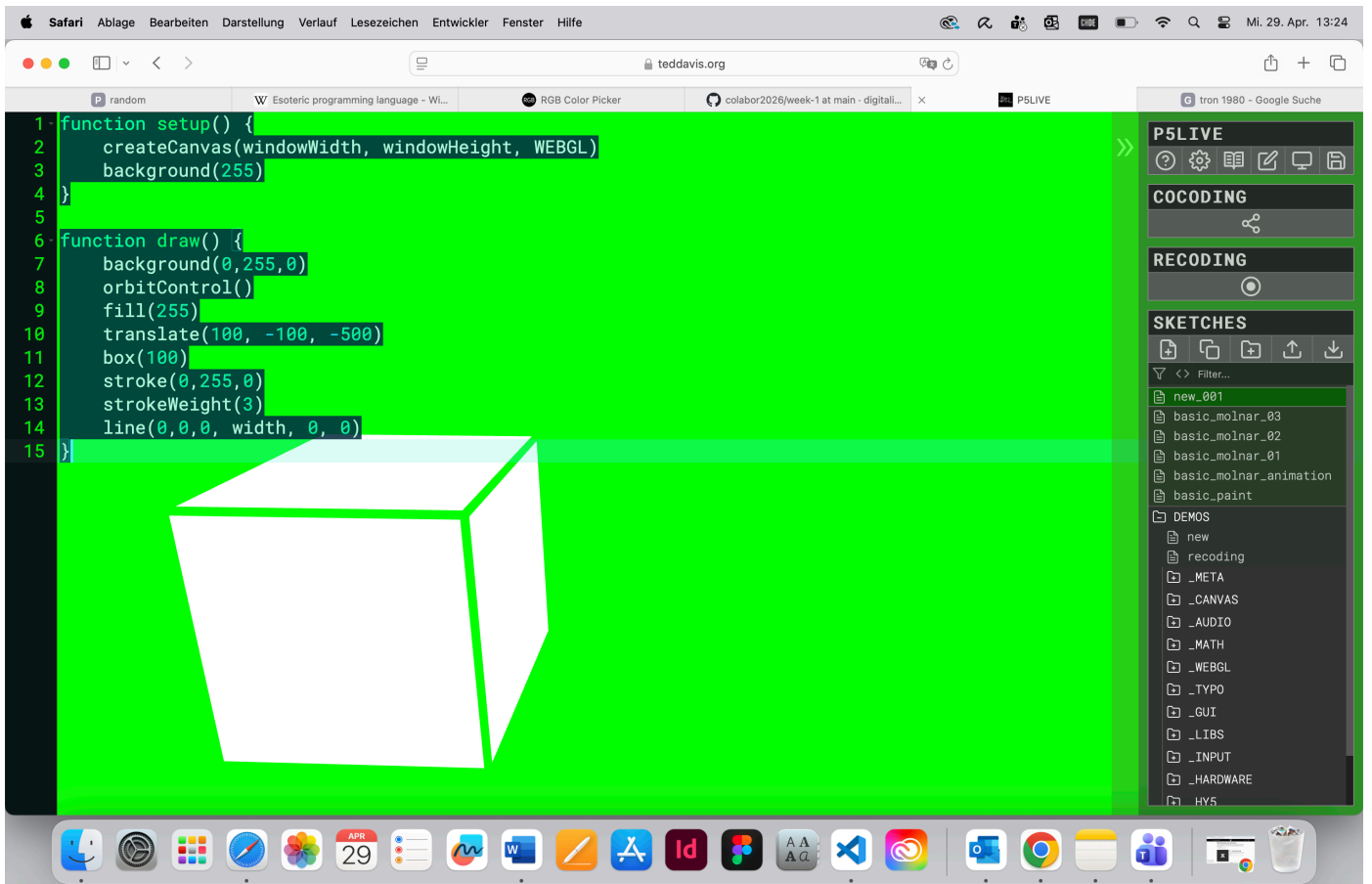
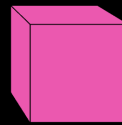
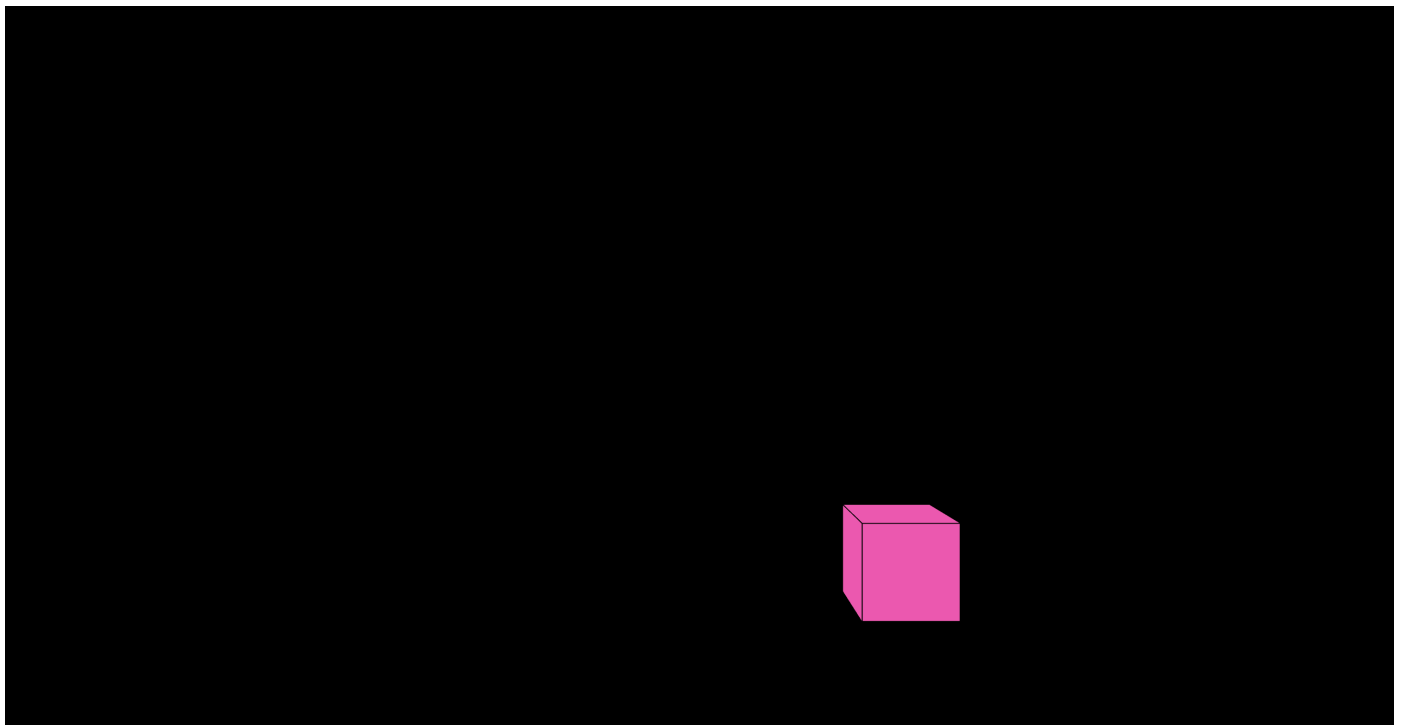
  x = x + xspeed;
  y = y + yspeed;
  z = z + zspeed;

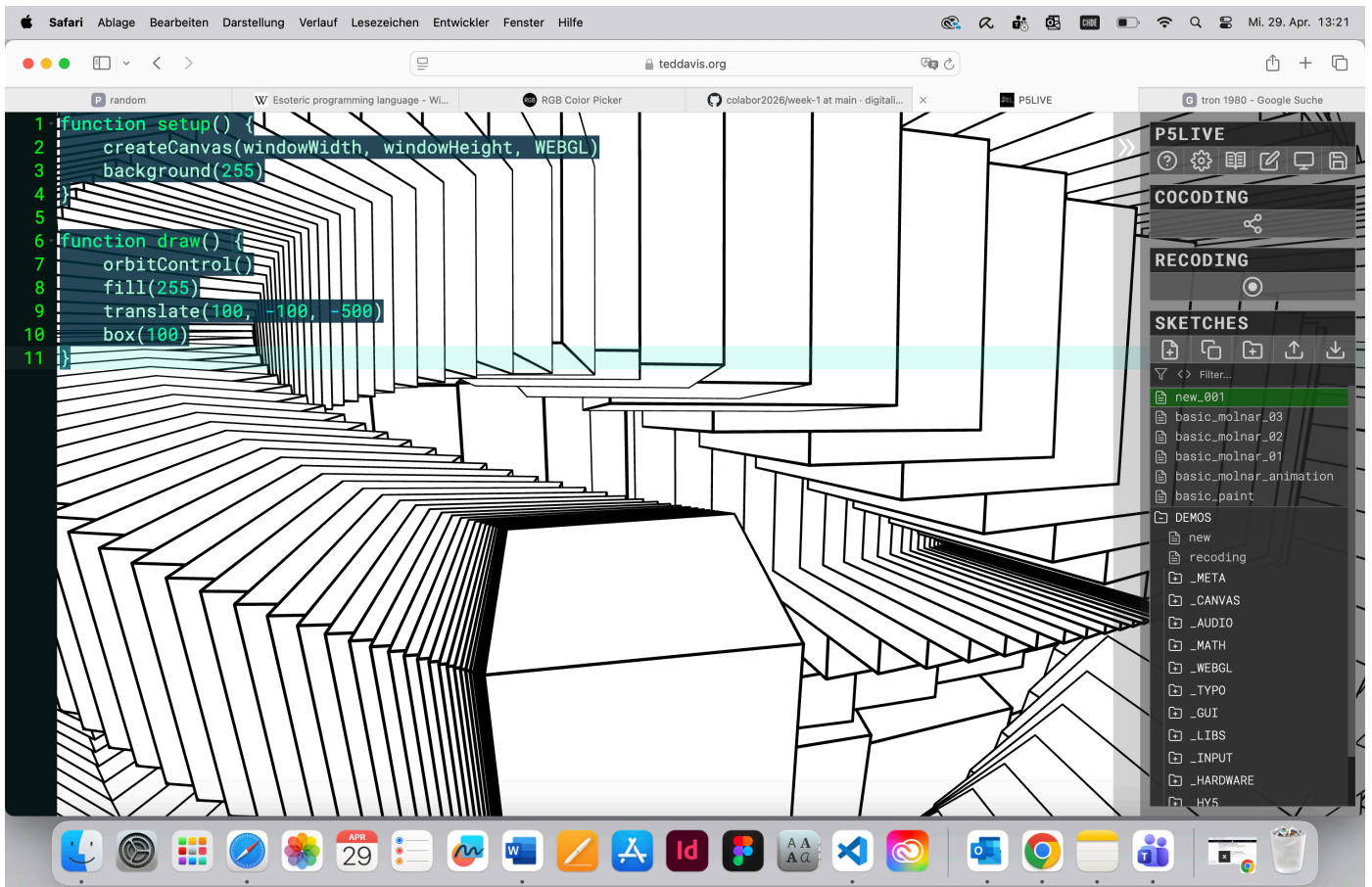
  if (x >= width) {
    xspeed = -xspeed;
    x = width;
    pickcolor();
  } else if (x <= 0) {
    xspeed = -xspeed;
    x = 0;
    pickcolor();
  }

  if (z >= 0) {
    zspeed = -zspeed;
    z = 0;
    pickcolor();
  } else if (z <= -depth) {
    zspeed = -zspeed;
    z = -depth;
    pickcolor();
  }

  if (y >= height) {
    yspeed = -yspeed;
    y = height;
    pickcolor();
  } else if (y <= 0) {
    yspeed = -yspeed;
    y = 0;
    pickcolor();
  }
}

```





## Day 4

Today we worked with VCV Rack 2

CHAOS !!! ⚡



